AI Plagiarism Checker
for Academic Reports

Uni 201 – Information Technology

Supervised by: [Salah Ali Ahmed]

Presented by:
- Ahmed Ashraf [PPTX]
- Ahmed Abdel-Tawab
- Esraa Khaled
- Arwa Nasreldin
- Shahd Arafa

Uni 201

INFORMATION TECHNOLOGY

Academic dishonesty is difficult to catch manually







Academic dishonesty: copying text in lab reports and assignments is a growing concern.

Manual review limits: checking reports manually takes too much time and effort.

The need: an automated IT tool to support original work and academic integrity.

The project turns similarity checking into an instant task



Automation

detect text similarity automatically.



Instant analysis

provide similarity percentages in seconds.



Accessibility

offer an easy-to-use interface for students and instructors.

From two documents to one similarity score



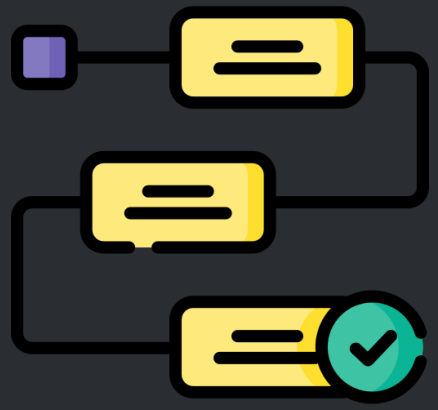
01 Input

upload or paste two text documents.



02 Text processing

clean the text and convert it into numerical vectors.



03 Similarity calculation

compare the vectors with a mathematical metric.



04 Output

display the similarity percentage instantly.

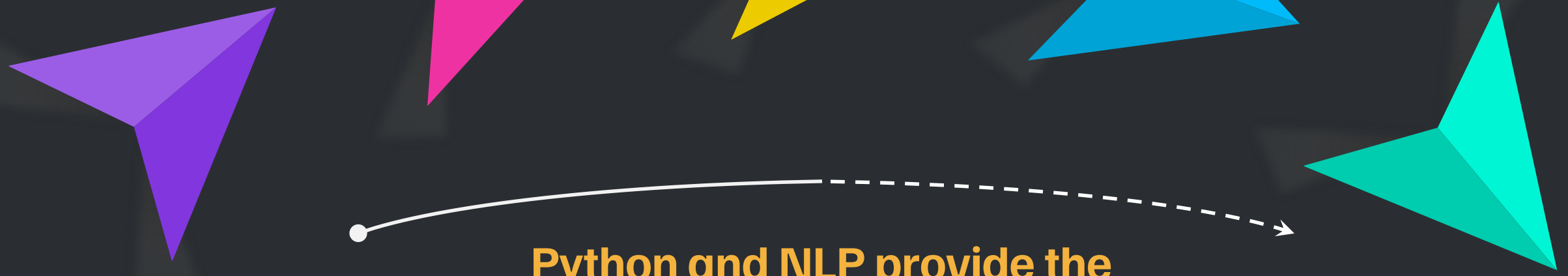
Programming language: Python 3.x

NLP technique: TF-IDF Vectorizer (Term Frequency–Inverse Document Frequency)

Similarity metric: Cosine Similarity

Core library: scikit-learn

User interface: Python Tkinter GUI



Python and NLP provide the technical foundation

Agile keeps development focused and testable



Planning

define requirements and scope.



Design

map the algorithm and simple user interface.



Implementation

write the Python similarity script.



Testing

validate accuracy using sample text files.

TF-IDF creates vectors; cosine similarity compares them

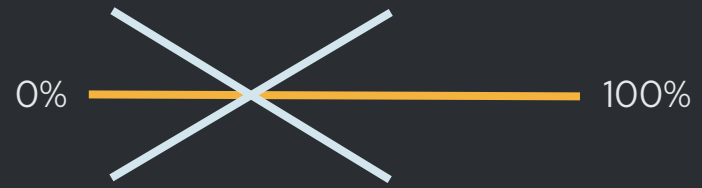
TF-IDF vectorization: convert words into mathematical values based on word frequency.

Cosine similarity: measure the angle between two text vectors in multi-dimensional space.

Formula: Similarity = (A · B) / (||A|| ||B||)

$$\text{Similarity} = \frac{A \cdot B}{\|A\| \|B\|}$$

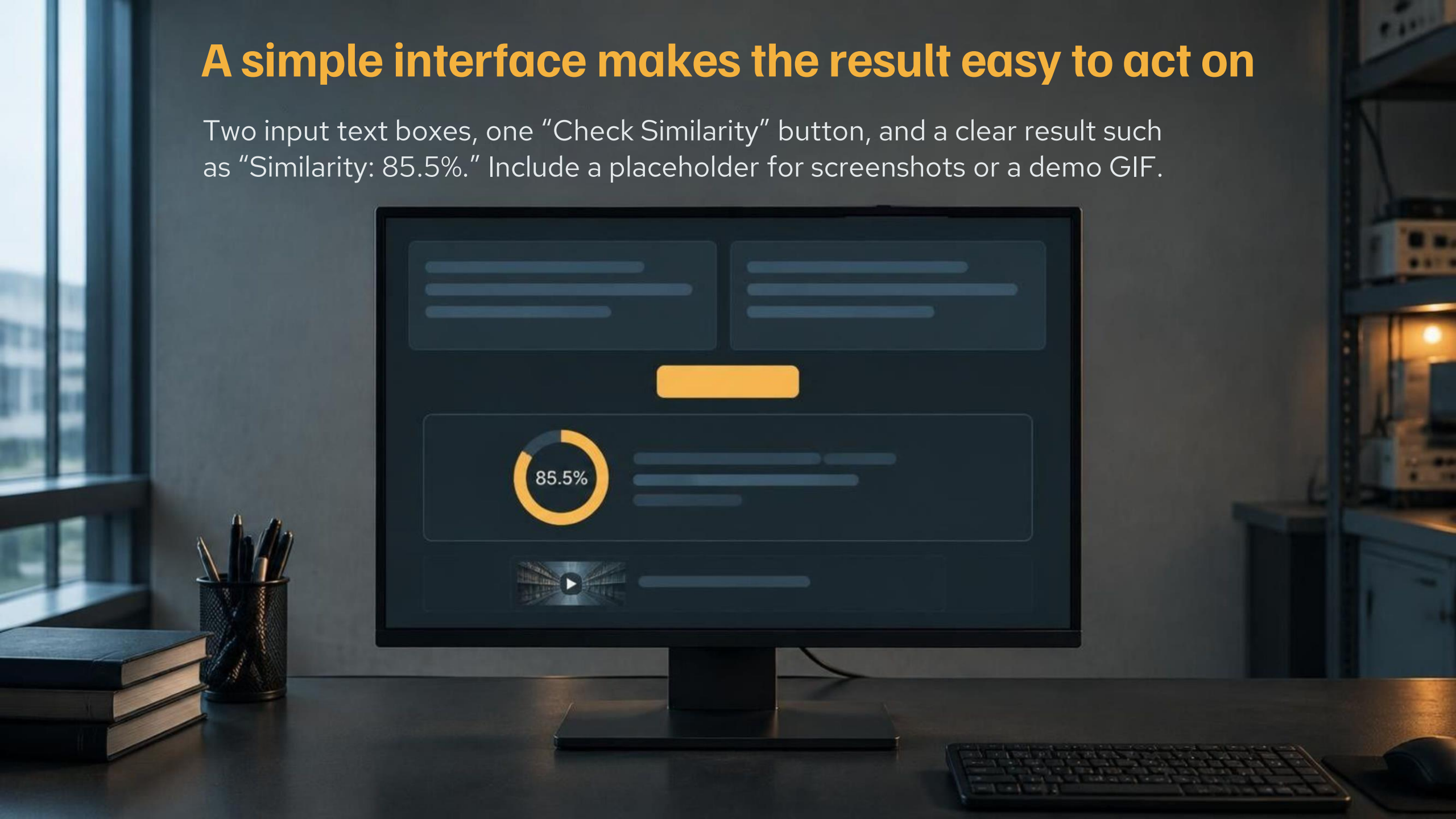
Result: 0% means unique; 100% means identical.



- 0% means unique
- 100% means identical.

A simple interface makes the result easy to act on

Two input text boxes, one “Check Similarity” button, and a clear result such as “Similarity: 85.5%.” Include a placeholder for screenshots or a demo GIF.



Fast, local, and privacy-friendly



Fast and lightweight: runs locally in seconds without heavy resources.



No external APIs: free, offline, and privacy-friendly.



Educational use: suitable for faculty assignments, lab reports, and research drafts.

A practical first step toward academic integrity

Future work: support PDF and Word document uploads.

Future work: add multi-language support, including Arabic text analysis.

Future work: integrate a web-based dashboard.

Conclusion: a simple, powerful IT solution to promote academic integrity.

References

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Thank You